

Key: The Ballots With Nothing On Them

84-355 Democracy's Data | Residual votes | Instructor copy — do not distribute

Jonathan Cervas

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All code is given to students. Grade the interpretation.

Structure, as of the rebuild. The brief (`residual-votes-brief.Rmd`) is now a chapter rather than a parallel copy of the lab: prose organised by argument, under idea headings, with four numbered figures and no numbered steps. The lab (`residual-votes.Rmd`) still runs Steps 1-10 with the code exposed. **Students should read the brief first. Step numbers below refer to the lab;** the chapter is cited by figure number or section title.

Requires EAVS and the Session 1 returns lab, and pays off the geography lab directly. Best taught after all three.

This is the only lab in the course where students construct a measure nobody publishes, and then use arithmetic to audit the sources it came from. Say that at the start — it is a different activity from every other session.

Every number below was run against the data files.

The session in one sentence

One ballot in a hundred carries no presidential vote, and the subtraction that tells you so also catches 103 counties where two federal datasets contradict each other outright.

The findings

```
c(counties_usable = nrow(u),
  median_rate     = round(median(u$residual_rate), 2),
  aggregate_rate  = round(100 * (sum(u$ballots) - sum(u$total_votes)) / sum(u$ballots), 2),
  impossible      = nrow(an),
  unusable_states = nrow(un))
```

```
#> counties_usable    median_rate aggregate_rate    impossible unusable_states
#>          2803.00           0.93           1.39          103.00           3.00
```

- **Median county residual rate 0.93%**, quartiles 0.64% and 1.39%.
- **103 counties** report more presidential votes than ballots cast.
- **Three states unusable**: Mississippi (all exact zeros), DC (87.7%), Vermont.

The three things to get said

1. The measure exists because of Florida 2000. Punch cards produced residual rates several times higher than optical scan, and the finding — from exactly this subtraction — is why they are gone. **A number nobody published turned out to be the one that mattered**, and building it is a demonstration that useful measures are sometimes lying around unassembled.

2. The floor is the audit. A residual vote cannot be negative. That single constraint converts a measurement into a **test of the sources**, and it is the only lab in the course where the data can prove itself wrong. Make sure students see the difference between “this county has an unusual value” and “this county has an impossible one.”

3. Three states fail in three different ways, and one is the geography lab. Mississippi answered a different question. DC is a unit mismatch — EAVS reports wards, the returns do not. **Point at DC explicitly and connect it back.** These were found by the build from the data itself, not from a list of known problems.

The line for the board: the subtraction measures the ballots, and the leftovers measure the bookkeeping.

Walking the steps

Steps	What they do	Min	Watch for
1-3	Two sources that were never meant to meet; one joined row; the township problem	8	Flag the nesting assumption here — Step 6 is where it detonates.
4	The distribution: median 0.93%	5	One ballot in a hundred. Make them convert it to votes in a close state.
5-7	The floor, the 103 impossible counties, and the other tail	14	The heart of the session. A measure that cannot be negative is a test of its own sources.
8-9	Three states where it breaks; what survives	8	Mississippi’s exact 0.000% everywhere is the one to linger on.
10	What you can say about the measure and about the sources	6	The finding is about the data system, not about voters.

Step 5 is the pivot and it is one sentence: a residual vote cannot be negative. Everything after it follows. Write it on the board before running Step 6.

If you are short on time, protect the pivot and the closing step. The numbers are what make the argument concrete; the sequence is the argument.

Option A — Which source is wrong?

Verified output

```
c(worst_gap_county = an$county_name[which.max(an$gap)],
  state = an$state_name[which.max(an$gap)],
  extra_votes = max(an$gap),
  median_gap = median(an$gap),
  gaps_of_10_or_fewer = sum(an$gap <= 10))

#>      worst_gap_county      state      extra_votes      median_gap
#>      "Cook County"      "Illinois"      "943822"      "112"
#> gaps_of_10_or_fewer
#>      "12"

head(an[order(-an$gap), c("state_name", "county_name", "ballots", "total_votes", "gap")], 5)

#>   state_name   county_name ballots total_votes   gap
#> 4  Illinois   Cook County  1112978    2056800 943822
#> 7  Missouri   Jackson County  196230     317702 121472
#> 5  Illinois   Winnebago County   70393     121048  50655
#> 6  Illinois   McLean County    51973      86168  34195
#> 2  Illinois   Knox County    10741      22175  11434
```

Cook County, Illinois: 943,822 more presidential votes than ballots cast — 85% of the reported ballots. The median anomaly is 112 votes; only 12 of the 103 are within 10 votes.

3/4: Picks counties, observes the size of the gaps, says the data cannot settle which source erred.

4/4: Separates the two populations. **Twelve counties differ by ten votes or fewer** — plausibly a late-counted provisional or an amended return, and not very interesting. **Cook County differs by nearly a million**, which is not an error of that kind at all.

4/4, best — and this is the answer to steer toward: diagnoses Cook County. **Chicago runs its own elections, separate from suburban Cook County**, so EAVS carries two jurisdictions where the returns file carries one county. Summing by FIPS prefix should catch both; if Chicago's rows are coded differently, the county's ballots are undercounted by roughly the size of Chicago. **A student who reasons from “who administers elections here” to “what the join must have missed” has used the geography lab exactly as intended.**

Option B — The state table

Verified output

```
agg <- tapply(seq_len(nrow(u)), u$state_name,
              function(i) 100 * (sum(u$ballots[i]) - sum(u$total_votes[i])) / sum(u$ballots[i]))
med <- tapply(u$residual_rate, u$state_name, median)
d <- data.frame(state = names(med), median = round(as.vector(med), 2),
               aggregate = round(as.vector(agg), 2))
d$gap <- round(d$aggregate - d$median, 2)
head(d[order(-abs(d$gap)), ], 6)
```

```
#>      state median aggregate  gap
#> 22 Missouri   1.36      5.26 3.90
#>  2  Arizona   1.04      2.55 1.51
#> 41   Utah     1.75      2.68 0.93
#> 11 Illinois   1.11      1.98 0.87
#> 29 New York   1.87      2.32 0.45
#> 12  Indiana   1.34      1.78 0.44
```

Missouri: 1.36% by median, 5.26% in aggregate.

3/4: Notes the two differ and that aggregate is vote-weighted.

4/4: Explains the direction. The aggregate is a ratio of sums, so **a handful of large counties with bad numbers drags it up**, while the median is immune. Missouri’s median says its typical county is unremarkable; its aggregate says several big ones are broken.

4/4, best: Says which to use and when — **the aggregate is the right estimate of “what share of ballots statewide were residual” if the data are clean, and the median is the better description of typical county performance when they are not.** Choosing between them requires knowing whether you trust the tails, which is the question the whole lab is about.

Option C — Set a filter

Verified output

```
for (cut in c(3, 5, 10, 100)) {
  k <- u$residual_rate <= cut
  cat(sprintf(" drop rate > %3d%%  keeps %4d  national %.2f%%\n",
              cut, sum(k), 100 * (sum(u$ballots[k]) - sum(u$total_votes[k])) / sum(u$ballots[k])))
}
```

```
#> drop rate > 3%  keeps 2697  national 1.20%
#> drop rate > 5%  keeps 2742  national 1.27%
#> drop rate > 10% keeps 2752  national 1.32%
#> drop rate > 100% keeps 2803  national 1.39%
```

```
for (m in c(0, 1000, 5000)) {
  k <- u$ballots >= m
  cat(sprintf(" ballots >= %5d keeps %4d national %.2f%%\n",
              m, sum(k), 100 * (sum(u$ballots[k]) - sum(u$total_votes[k])) / sum(u$ballots[k])))
}
```

```
#> ballots >= 0 keeps 2803 national 1.39%
#> ballots >= 1000 keeps 2722 national 1.39%
#> ballots >= 5000 keeps 2139 national 1.39%
```

Two rules, two completely different outcomes. Trimming by rate moves the national figure from 1.39% to 1.20%. **Trimming by county size does nothing at all** — 1.39% whether you keep every county or only those with 5,000+ ballots.

3/4: Picks a rule, reports how many counties it removes and how the figure moves.

4/4: Notices the size filter is inert, and says why that is informative: **the bad rows are not small counties**. The instinct that noise lives in small places is wrong here — Cook County is the worst offender in the file.

4/4, best: Sees the circularity in the rate filter. **Excluding counties because their residual rate is implausible uses the outcome to select the sample**, which will always tighten the estimate and cannot distinguish bad data from genuinely bad ballot design. A student who says the honest move is to report both figures and show the excluded rows — which is what the lab does — has arrived at the professional standard.

Grading

Four-point scale from the syllabus.

- **Never penalise code.** It is your code.
- **The move that earns a 4** is distinguishing *impossible* from *unusual*, and treating the impossible ones as evidence about the sources.
- **A 2** is reporting 0.93% as the answer with no engagement with the 103.
- **Also a 2:** “the data is bad so we can’t say anything.” 2,803 counties behaved.
- **Reward anyone who diagnoses a specific anomaly** rather than listing them.

Anticipated questions

- “Which number is the real residual rate?” — Median 0.93% for a typical county; 1.39% across the usable states by volume. Both, with the caveat.
- “Is 1% bad?” — Depends entirely on the margin. In 2000 Florida it was decisive; in a 20-point state it is invisible. **Refuse to answer without a comparison.**
- “Why doesn’t the EAC just fix this?” — EAVS reports what jurisdictions submit, and the county returns come from state canvasses. **No agency owns the comparison**, which is why nobody had made it.
- “Could students report an error?” — Genuinely yes. The EAC accepts corrections and the anomalies file is a list of candidates. Worth mentioning: the work is real.

Connections

- **EAVS** — the source of the ballots figure and the same self-report problem.
- **Session 1 returns** — the other half, and the first “two sources disagree” lesson.
- **Census geography** — DC’s failure is a unit mismatch; Cook County’s is an administrative one.
- **Independent expenditures** — the other lab where impossible values exposed what nobody had checked.